St. Francis Institute of Technology (Engineering College)

An Autonomous Institute, Affiliated to University of Mumbai

NAAC A+ Accredited | CMPN, EXTC, INFT NBA Accredited | ISO 9001:2015 Certified

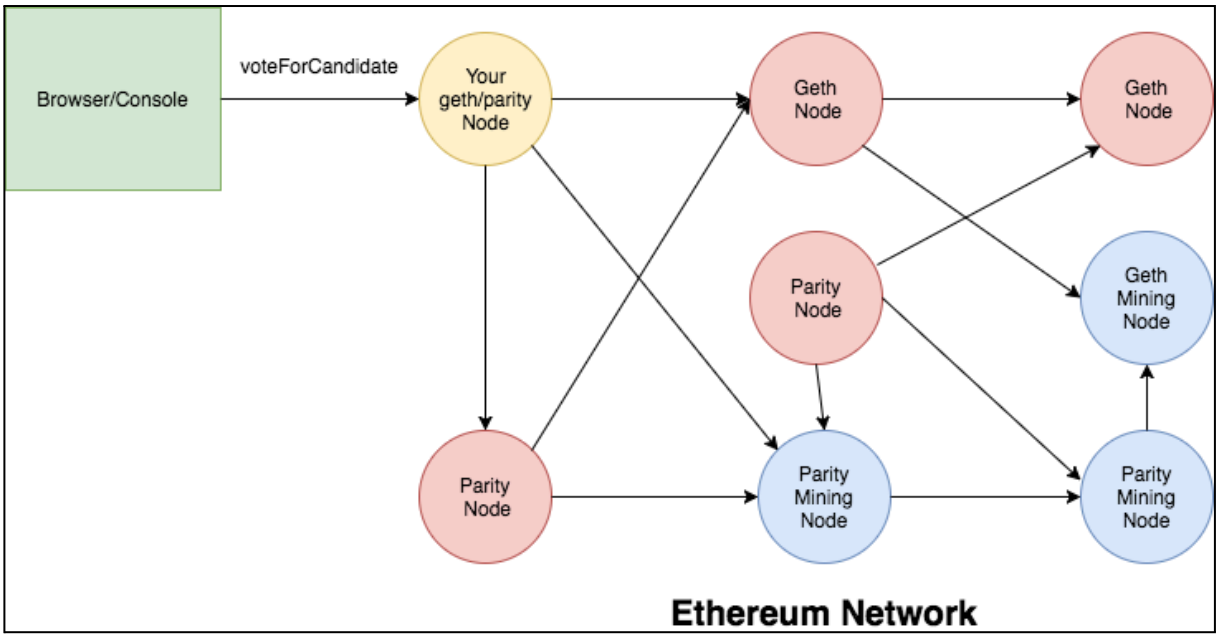
Class: BEIT – A Sem – VIII Subject: Blockchain & DLT

Module 3 - Permissionless Blockchain: Ethereum

Assignment

Q4. What are different types of Ethereum transactions? Explain each of them with a simple example.

ANS:



In Ethereum, a transaction is a digitally signed message sent from one account to another to transfer Ether or execute operations on the blockchain. Transactions are initiated by externally owned accounts and are validated by the network before being permanently recorded

on the blockchain. They can be used not only for transferring cryptocurrency but also for deploying and interacting with smart contracts.

1) Ether Transfer Transaction

1. Used to transfer Ether (ETH) from one account to another.
2. It is the simplest type of transaction.
3. Does not involve any smart contract.
4. Requires a small gas fee for processing.
5. **Example:** User A sends 1 ETH to User B.

2) Contract Creation Transaction

1. Used to deploy a new smart contract on the Ethereum blockchain.
2. Does not have a receiver address.
3. Contains contract code (bytecode).
4. Generates a new contract address after deployment.
5. **Example:** A developer deploys a voting or donation smart contract.

3) Contract Interaction Transaction (Message Call)

1. Used to interact with an already deployed smart contract.
2. Calls functions defined inside the contract.
3. Can read or modify data stored in the contract.
4. Requires gas if it changes the contract state.
5. **Example:** A user sends ETH to a donation contract or calls a function like “vote()”.

Q6. What is solidity? Describe the structure and purpose of simple solidity smart contract including function, state variables and modifiers.

ANS:

1. Solidity is a high-level, object-oriented programming language used to write smart contracts.
2. It is mainly used on the Ethereum blockchain.
3. It is statically typed and supports features like inheritance and libraries.
4. Its syntax is similar to JavaScript and C++.

Structure of a Simple Solidity Smart Contract

1) State Variables

1. State variables are stored permanently on the blockchain.
2. They are declared inside the contract but outside functions.
3. They store data like balance, owner address, or counters.
4. They can have visibility such as public or private.
   Example: uint public count;

2) Functions

1. Functions define the behavior of the smart contract.
2. They can read or modify state variables.
3. They can be public, private, internal, or external.
4. Gas is required if they modify the contract state.

Example: function increment() public { count++; }

3) Modifiers

1. Modifiers are used to control access to functions.
2. They check conditions before executing a function.
3. They improve security and code reusability.
4. Written using the modifier keyword.
   Example: modifier onlyOwner { require(msg.sender == owner); _; }

Purpose of a Simple Smart Contract

1. To automate agreements without intermediaries.
2. To ensure transparency and trust.
3. To securely store and manage data.
4. To execute predefined rules automatically.

Q8. What is gas in Ethereum transactions, why it is important?

ANS:

1. Gas is the **fee required to perform any operation** on the Ethereum blockchain.
2. It measures the **computational effort** needed to execute transactions or smart contracts.
3. Gas is paid in Ether (ETH), usually in smaller units like **Gwei**.

Why Gas is Important

1. It **prevents network abuse and spam**
   - Users must pay for every transaction, so unnecessary or malicious transactions are reduced.

2. It **compensates miners/validators**
    - Gas fees act as a reward for miners or validators who process and validate transactions.
3. It ensures **efficient use of resources**
    - More complex operations require more gas, encouraging optimized code.
4. It helps in **transaction prioritization**
    - Transactions with higher gas fees are processed faster by the network.
5. It provides **security to the network**
    - Prevents infinite loops or heavy computations in smart contracts by requiring gas for each step.

Example

1. Sending ETH from one account to another requires a fixed amount of gas (e.g., 21,000 gas).

Conclusion

1. Gas is essential for running transactions on Ethereum.
2. It maintains security, efficiency, and proper functioning of the blockchain network.

Ms. Shraddha Sandimani Dr. Prachi Raut Subject Incharge (HOD- INFT)